



AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0036



Agricultural Machinery – Brush Cutter HONDA UMK435T

Test Report Number	2023 – 0036
Date of Issuance	February 14, 2023
Valid Until	February 14, 2028
Agricultural Machinery	Brush Cutter
Type	Strap type, Straight shaft, Portable
Brand	HONDA
Model	UMK435T
Test Requested by	HONDA PHILIPPINES INC. Lot 34 Phase 1B, First Philippine Industrial Park, Tanauan City, Batangas

SUMMARY

Performance Criteria	Manufacturer's Specification	AMTEC Test
Actual Field Capacity, ha/h	ND ^a	0.102
Average Width of Cut, mm	ND ^a	2031
Fuel Consumption, L/h	ND ^a	0.818

^a Not specified by the requesting party.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English, French, Spanish
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	2016 – 0007, 2018 – 0435
3. Manufacturer	THAI HONDA MFG. CO., LTD. 410 Thanon Liap Khlong Lam Ko Phai, Lam Prathew, Lat Krabang, Bangkok 10520, Thailand
4. Prime mover used	1.0 kW HONDA GX35 Gasoline Engine
5. Number of operators	One
6. Handling and stability at operation	Stable operation was observed during the test operation.
7. Adjusting and repairing of parts	The length of straps can be adjusted. The blade is also replaceable.
8. Cleaning of parts	All parts are accessible for cleaning.
9. Safety features	The machine is equipped with a deflector. Warning stickers are also provided.
10. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	None
11. Other remarks	No breakdown occurred during the test operation.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	February 1, 2023
Location of Test	AMTEC Compound, UPLB, College, Los Baños, Laguna 14°9'49.80" N, 121°14'53.80" E
Items Tested	Operator's Manual, Speed of Operation, Width of Cut, Height of Cut, Fuel Consumption Rate, Theoretical and Actual Field Capacity, Operator's Pulse Rate, and Noise Level

METHODS OF TEST

The brush cutter was tested following the AMTEC Methods of Test.

DESCRIPTION OF THE MACHINE

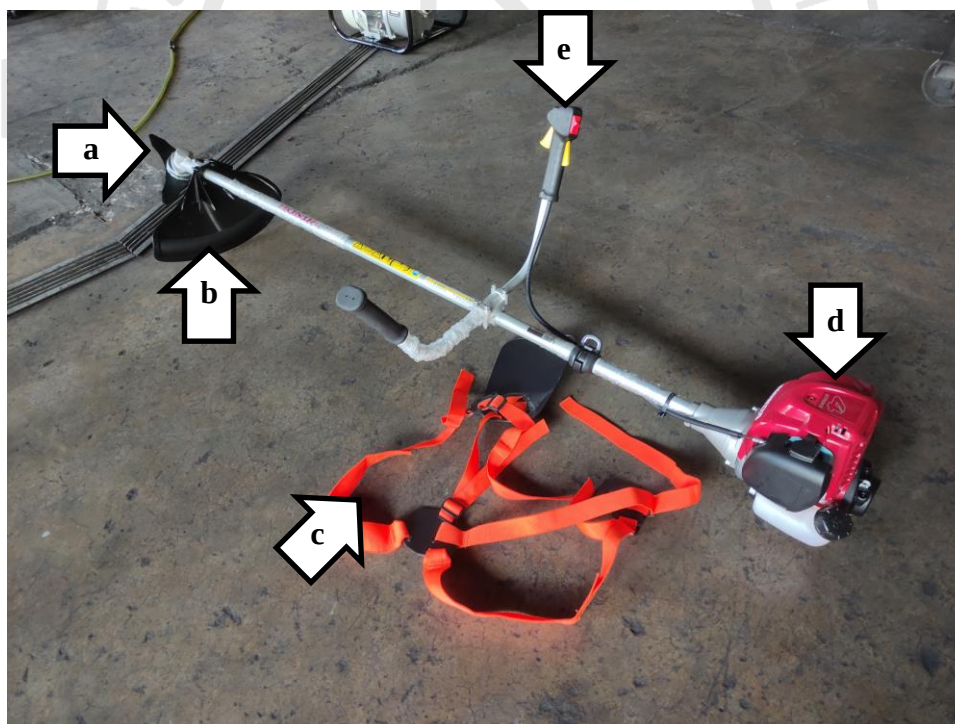


Figure 1. Parts of the HONDA UMK435T Brush Cutter powered by 1.0 kW HONDA GX35 Gasoline Engine include the following: (a) Cutting blade, (b) Safety cover/Deflector, (c) Harness/Strap, (d) Prime mover, and (e) Handle with throttle lever.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Overall dimensions, mm		
1.1 Length	1901	1905
1.2 Width	640	640
1.3 Height	420	465
2 Dry weight, kg	7.7	8.0 ^b
3 Cutting blade		
3.1 Dimensions, D × t, mm	255 × ND	250 × 4.20
3.2 Material	ND	S-5 ^c
3.3 Configuration	3-teeth blade	3-teeth blade
4 Safety cover/Deflector		
4.1 Dimensions, L × W × t, mm	ND	380 × 73 × 2.87
4.2 Material	ND	Polymer
4.3 Location	ND	Near the blade
5 Backpack/Shoulder strap		
5.1 Length, mm	ND	1390
5.2 Width, mm	ND	50
5.3 Thickness, mm	ND	2.05
5.4 Material	ND	Non-absorbent fabric
5.5 Strap pad	NA	NA
6 Engine		
6.1 Brand	HONDA	HONDA
6.2 Model	GX35	GX35
6.3 Type	Four-stroke, one cylinder, OHC	Four-stroke cycle, single cylinder, OHC
6.4 Make	Thailand	Thailand
6.5 Serial number	ND	ND
6.6 Rated output power, kW	1.0	NM
6.7 Fuel system		
6.7.1 Type of fuel feed	ND	Gravity
6.7.2 Type of fuel used	Gasoline	Gasoline
6.7.3 Tank capacity, L	0.63	0.686
6.8 Cooling system	Forced air	Air-cooled
6.9 Starting system	ND	Rope recoil
6.10 Exhaust system	Muffler	Muffler

^a The data were obtained from the operator's manual provided by the requesting party.

^b Includes weight of the engine oil.

^c Determined using the Bruker S1 Titan 800 X-ray Fluorescence Analyzer; See Table 1 for elemental composition.

Note: NA – Not Applicable, ND – No Data, NM – Not Measured

RESULTS

Item		Data
1	Test conditions	
1.1	Type of field operation	Grass cutting
1.2	Condition of field	
1.2.1	Location	AMTEC Compound, UPLB, College, Los Baños, Laguna
1.3	Dimension of field	
1.3.1	Length, m	22
1.3.2	Width, m	11
1.3.3	Area, m ²	242
1.4	Weed density	High
1.5	Average grass height, mm	1444
2	Field performance	
2.1	Traveling/operating speed, kph	0.542
2.2	Average width of cut, mm	2031
2.3	Average height of cut, mm	271
2.4	Average engine speed during operation, rpm	5126
2.5	Actual field capacity	
2.5.1	ha/h	0.102
2.5.2	h/ha	9.83
2.6	Theoretical field capacity	
2.6.1	ha/h	0.110
2.6.2	h/ha	9.19
2.7	Field efficiency, %	93.24
2.8	Fuel consumption	
2.8.1	L/h	0.818
2.8.2	L/ha	8.01
2.9	Noise at operator's ear level, dB(A)	92.5
2.10	Operator's pulse rate, bpm	
2.10.1	Before operation	95
2.10.2	After operation	144
2.11	Operator's blood pressure, systolic/diastolic	
2.11.1	Before operation	116/92
2.11.2	After operation	125/89
2.12	Assessment of workload ^a	High

^a Based-on Table 5.3 of FAO Agricultural Services Bulletin 110: Testing and Evaluation of Agricultural Machinery and Equipment (1994).

RESULTS (Cont'n)

Table 1. Elemental composition of the blade of HONDA UMK435T Brush Cutter

Element	Percent Composition, %^a
Silicon	3.168
Phosphorus	0.563
Sulfur	0.091
Vanadium	0.012
Chromium	0.210
Manganese	0.473
Iron	94.172
Nickel	0.975
Copper	0.141
Molybdenum	0.062
Tin	0.093

^a Determined using the Bruker S1 Titan 800 X-ray Fluorescence Analyzer

OBSERVATIONS



Figure 2. Before operation.



Figure 3. After operation.

OBSERVATIONS (Cont'n)

Figure 4. Markings and warning stickers of the HONDA UMK435T Brush Cutter.

Tested by:


PAUL JOHN S. DIZON

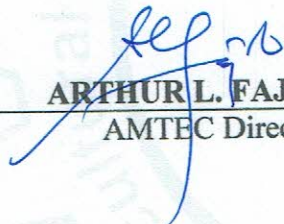
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Test Engineer

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FEB 16 2023

Date

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